



grant adams - NOAA Federal <grant.adams@noaa.gov>

Hybrid Fishery Data

12 messages

Jackson, Tyler M (DFG) <tyler.jackson@alaska.gov>

Thu, Mar 5, 2026 at 4:48 PM

To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>, grant adams - NOAA Federal <grant.adams@noaa.gov>

Cc: "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Hi Buck, Cody, Grant,

Still working on some fine details of this hybrid data, but as promised I wanted to get you what I have available by end of week, which should suffice for what you're being asked for in May. Attached are total catch and total/retained size composition data of Chionoecetes hybrids. There's a bit of hand wringing involved, so stay with me here...

- Hybrids can be recorded 3 ways by observer and dockside samplers 1) opilio-type, 2) bairdi-type, 3) unspecified. The 'types' refer to the regulatory ID specifically. For instance, a bairdi-type hybrid would have red eyes and an M-shaped epistome, but otherwise have some combination of carapace, spines, rostrum shape that would lead the recorder to ID as a hybrid. Likewise, an opilio-type would not have red eyes and M shaped epistome, yet would have secondary characteristics not consistent with a pure snow crab. **What I'm providing now is estimates with all three lumped together**, so purely what have been IDed as a biological hybrid irrespective of how they'd be counted in regulation. This is because there are many 'unspecified' hybrids mixed in throughout the timeseries.
- The calculated weight used in estimation of total catch uses the same length-weight parameters as snow crab. Opilio-type is much more common than bairdi-type and as long as we are lumping all hybrids together, this is the most straightforward approach.
- These data are hybrid only. So if you wanted to simply treat them in your model as the parent species, you'd sum these to your current values.
- I do not have retained catch apportioned by hybrid and parent species yet. The retained catch numbers you have are complete in the sense that they include all landed crab. But if for instance we want to be specific enough to break down the snow crab retained catch by which portion was actually hybrids, I've got to know how to deal the different 'types' and specifically the unspecified hybrids, which we've not totally resolved yet.
- As usual, weight is in tonnes. Total catch estimates are broken down by females, sublegals, and legals (it's how the observer counts are recorded). Size composition are simply count of crab, total number measured is the sum by fishery. For Grant – if you haven't caught on total catch is total estimated retained + discards expanded from observer pots, not total mortality which we unfortunately call 'total catch' in the SAFE intros.
- ADF&G and NMFS aren't exactly in lock step when it comes to what number of characteristics equate to a positive hybrid ID. Something that should be part of discussion at the CPT meeting.

Let me know if you have any questions,

Tyler Jackson

Biometrician | ADF&G Commercial Fisheries

tyler.jackson@alaska.gov

(907) 486 - 1861

3 attachments

-  **hybrid_total_catch.csv**
39K
-  **hybrid_total_catch_comp.csv**
102K
-  **hybrid_retained_catch_comp.csv**
40K

grant adams - NOAA Federal <grant.adams@noaa.gov>

Fri, Mar 6, 2026 at 11:09 AM

To: "Jackson, Tyler M (DFG)" <tyler.jackson@alaska.gov>

Cc: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Hi Tyler,

Got the data, thanks for sending this over! I'm going to go through things with Cody next week and will ping you if we have any questions.

Thank you!

Grant

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William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Fri, Mar 6, 2026 at 11:42 AM

To: grant adams - NOAA Federal <grant.adams@noaa.gov>

Cc: "Jackson, Tyler M (DFG)" <tyler.jackson@alaska.gov>, cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Thanks Tyler! Will start on it next week as well.

Grant, let's all brainstorm together next week. Might as well have similar approaches.

William T. Stockhausen*Research Fishery Biologist, Alaska Fisheries Science Center*

NOAA Fisheries | U.S. Department of Commerce

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cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Tue, Mar 10, 2026 at 1:28 PM

To: William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>, "Jackson, Tyler M (DFG)" <tyler.jackson@alaska.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Hi Tyler!

We're getting ready to work through getting these data into the assessment and I had a question about what the retained catch for snow crab currently has in it. I seem to remember hybrids are counted as snow crab, so I'm guessing that they are in the retained biomass, but are they also in the size composition data for snow crab?

Also, are the numbers reported in the size composition scaled to the abundance caught at size? I think you were implying this when you wrote "So if you wanted to simply treat them in your model as the parent species, you'd sum these to your current values," but I wanted to make sure I was interpreting that correctly.

Thanks again for getting all of this together for us,

Cody and company

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Jackson, Tyler M (DFG) <tyler.jackson@alaska.gov>

Tue, Mar 10, 2026 at 2:10 PM

To: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

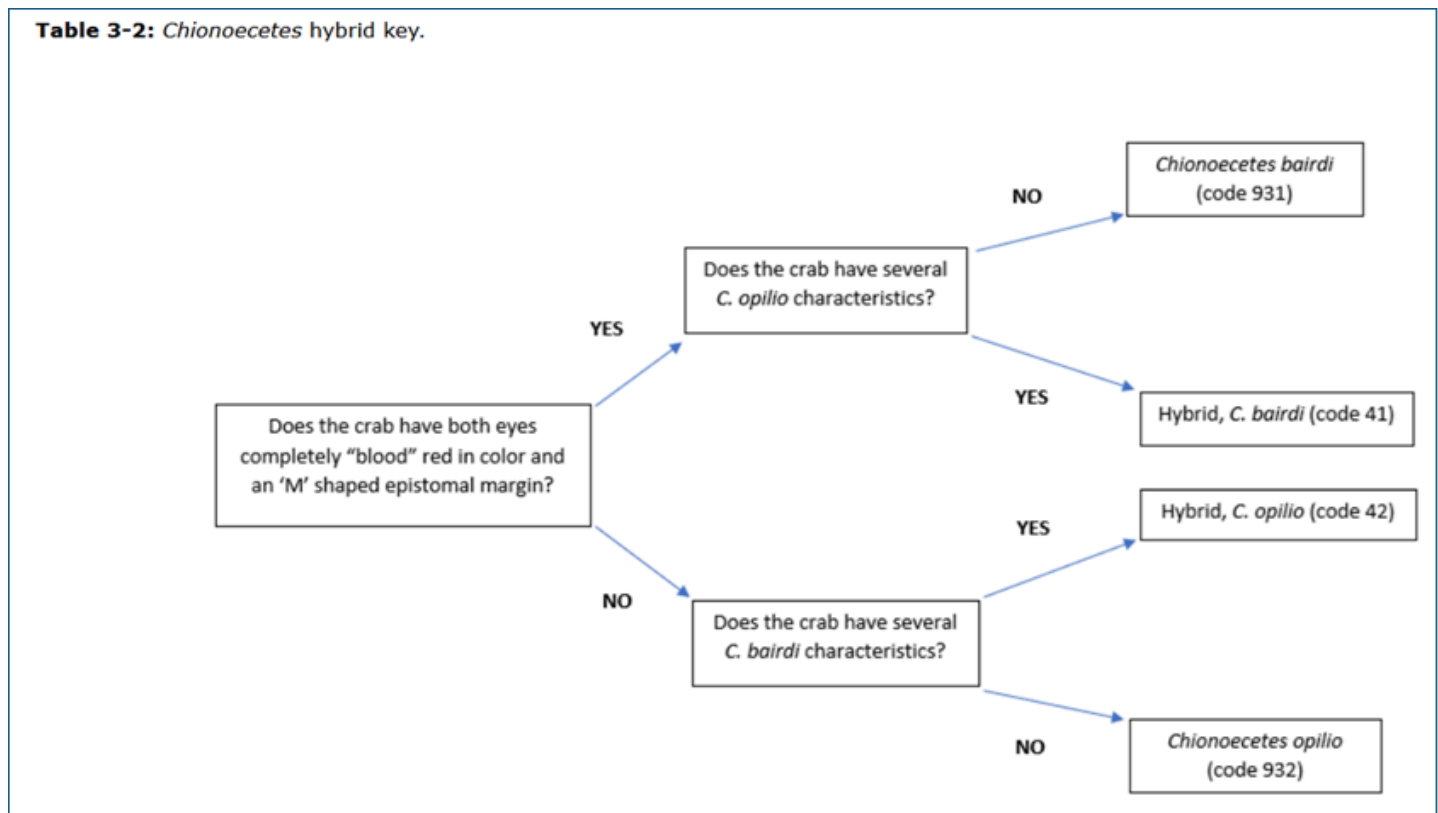
Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Answers:

1. Yes, there would be some portion of the snow crab retained catch numbers – that you currently have – that would be hybrids. To a much, much, lesser extent, this would also be true for Tanner crab. While the vast majority of hybrids do not meet the regulatory definition of a Tanner crab (red eyes, M shaped mouth), the data suggest a small handful apparently do. That's where the bairdi- and opilio- 'types' come in, I found the flow chart below very helpful.
2. Hybrids would not be in the size composition data that you previously had – those would be pure snow or Tanner crab only. This goes for the total catch files you previous had as well.
3. Size composition data aren't relative, just numbers at size. Total number measured by fishery is the total sum by fishery.
4. What I meant by summing data was that if you wanted, say, the overall size composition of the 2024 BSSC directed fishery (including both snow and hybrids) you'd simply sum the numbers at size from the species-specific data for that fishery. This would go for total catch too. I can't remember the specifics of how you were asked to set up these models, but I thought it was something like 'assuming hybrids counted as the parent species'...

Does that clear things up?

T



From: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>

Sent: Tuesday, March 10, 2026 12:29 PM

To: William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Cc: grant adams - NOAA Federal <grant.adams@noaa.gov>; Jackson, Tyler M (DFG) <tyler.jackson@alaska.gov>; Palof, Katie J (DFG) <katie.palof@alaska.gov>; Daly, Ben J (DFG) <ben.daly@alaska.gov>; Nichols, Ethan W (DFG) <ethan.nichols@alaska.gov>

Subject: Re: Hybrid Fishery Data

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William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Tue, Mar 31, 2026 at 9:48 AM

To: "Jackson, Tyler M (DFG)" <tyler.jackson@alaska.gov>

Cc: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, grant adams - NOAA Federal <grant.adams@noaa.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Hi Tyler,

Hope all is going well! Finally getting to the hybrid fishery data after struggling with Tanner crab and gmacs for too long. I was really surprised at the amount of hybrid catch in the 1997 snow crab fishery: ~18,000 tonnes legal males! Is that really correct? Wow!

Just want to make double sure on that. Thanks for confirming it, if you can!

Buck

William T. Stockhausen*Research Fishery Biologist, Alaska Fisheries Science Center*

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William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Tue, Mar 31, 2026 at 9:51 AM

To: "Jackson, Tyler M (DFG)" <tyler.jackson@alaska.gov>

Cc: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, grant adams - NOAA Federal <grant.adams@noaa.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Also, the weight units are metric tons? Thought "tonnes" was just Olde English for "tons" until Googling it just now.

William T. Stockhausen*Research Fishery Biologist, Alaska Fisheries Science Center*

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Jackson, Tyler M (DFG) <tyler.jackson@alaska.gov>

Tue, Mar 31, 2026 at 11:33 AM

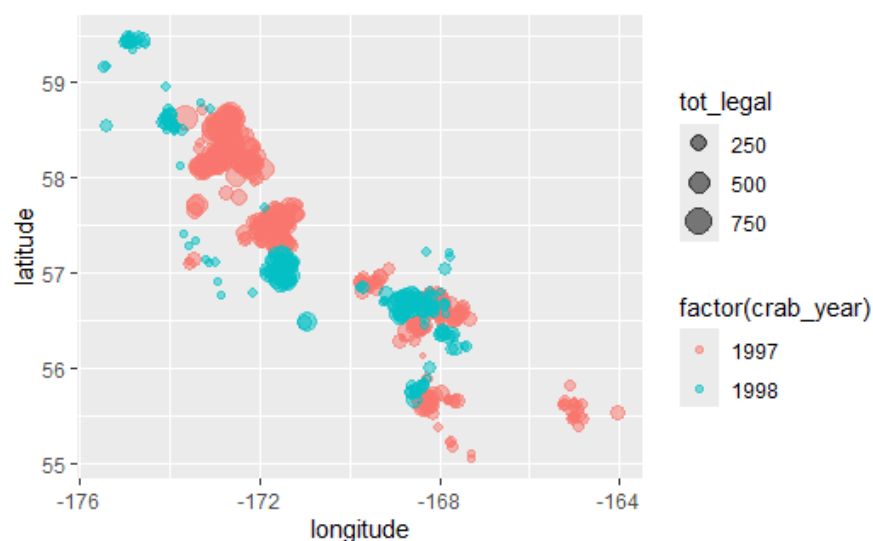
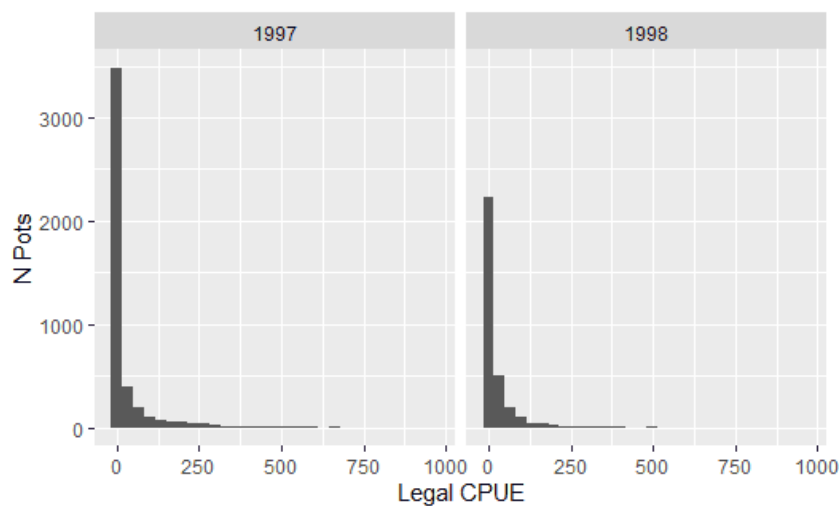
To: William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

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Yes metric! Probably should just start saying metric ton I guess.

Yea, good catch. It's a surprising amount, there are a couple years with really high catches. The math is correct – the data are maybe suspect. Observer hybrid data quality gets fishy-er the further back in time you go. There have been evolving ID criteria and stances on how things should be lumped vs split. 1997 is getting into that realm for sure.

I decided it would be best to hand off data as is for this go around and highlight the uncertainty as I present it. We can track down and decide what to do with some of these discrepancies if using this data in assessment becomes the norm. In that year, the really high hybrid CPUE pots are definitely fewer, but peppered around spatially and not from one observer or vessel. Similar estimate in 1998 (~15k t), but the spatial hot spots (CPUE > 100) don't really align.



Tyler

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William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Tue, Mar 31, 2026 at 1:07 PM

To: "Jackson, Tyler M (DFG)" <tyler.jackson@alaska.gov>

Cc: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, grant adams - NOAA Federal <grant.adams@noaa.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Thanks for the info, Tyler! Spatial plot is really interesting.

William T. Stockhausen

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William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Tue, Mar 31, 2026 at 3:43 PM

To: "Jackson, Tyler M (DFG)" <tyler.jackson@alaska.gov>

Cc: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, grant adams - NOAA Federal <grant.adams@noaa.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Hey again Tyler,

Referring back to your first email in this chain, you wrote:

"I do not have retained catch apportioned by hybrid and parent species yet. The retained catch numbers you have are complete in the sense that they include all landed crab. But if for instance we want to be specific enough to break down the snow crab retained catch by which portion was actually hybrids, I've got to know how to deal the different 'types' and specifically the unspecified hybrids, which we've not totally resolved yet"

If I'm understanding the above correctly, the retained catch abundance and biomass that I have for Tanner crab (both in the directed fisheries and incidentally retained in the snow and BBRKC fisheries) already includes hybrids, so I don't need to worry about any further aggregation to include retained hybrids. Is that correct?

Also, just for the record, what were the values of the length-weight relationships you used to get the hybrid weights?

Thanks again! (and sorry to have to keep bugging you)

William T. Stockhausen*Research Fishery Biologist, Alaska Fisheries Science Center*

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Jackson, Tyler M (DFG) <tyler.jackson@alaska.gov>

Tue, Mar 31, 2026 at 3:53 PM

To: William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Cc: cody szuwalski - NOAA Federal <cody.szuwalski@noaa.gov>, grant adams - NOAA Federal <grant.adams@noaa.gov>, "Palof, Katie J (DFG)" <katie.palof@alaska.gov>, "Daly, Ben J (DFG)" <ben.daly@alaska.gov>, "Nichols, Ethan W (DFG)" <ethan.nichols@alaska.gov>

Correct. Retained catch has always included hybrids, but as either snow or Tanner crab.

Calculated weight for all hybrids used the weight at size parameters of snow crab (the overwhelming majority are of the "opilio type")

Male: $\alpha = 0.000267$, $\beta = 3.097253$

Mature Females: $\alpha = 0.001158$, $\beta = 2.708793$

Immature Females: $\alpha = 0.001047$, $\beta = 2.708367$

T

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William Stockhausen - NOAA Federal <william.stockhausen@noaa.gov>

Tue, Mar 31, 2026 at 4:24 PM

To: "Jackson, Tyler M (DFG)" <tyler.jackson@alaska.gov>

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Thanks Tyler. Good we don't have to do any fiddling with retained catch abundance and biomass, at last until the CPT and SSC realizes we probably should have ideally been trying to exclude hybrids from that data before they went into the assessment models.

William T. Stockhausen

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